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A SINGLE PROBE TRANSIENT MOVING LINE SOURCE METHOD FOR DETERMINING THERMAL CONDUCTIVITY, THERMAL DIFFUSIVITY, AND SUPERFICIAL FLOW VELOCITY OF A POROUS MATERIAL WITH FLOW

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ABSTRACT

This paper presents a method to determine the effective thermal conductivity and thermal diffusivity of a porous material, as well as the superficial flow velocity of fluid flowing through the porous matrix using a single probe transient moving line source method. The method transforms the transient analytical solution for a moving line source using differentiation to produce three independent equations to solve for the three unknowns. Empirical data are presented from a laboratory scale test apparatus for three test cases with known properties and flow rates to validate the method. The method is then applied to field data from a standing column well used in ground source heat pump systems to obtain the thermal and flow properties of the ground formation. The properties are inserted into the transient analytical solution for a moving line source and superimposed over the empirical data showing agreement between the model and the data. The method is more accurate than traditional methods for estimating thermal properties when flow conditions are present, and the implementation of the method does not require any additional thermal data.

Keywords: Thermal properties, hydraulic flow, ground source heat pumps, volumetric flux

INTRODUCTION

Acquiring thermal properties of perfused materials is necessary for accurate design of systems and products in various industries such as food, agriculture, and ground source heat pump systems. [1-5] Many of the methods for acquiring thermal properties are standardized by the American Society for Testing and Materials (ASTM). The standards for the transient line source method, the guarded hot plate method, and the flash method are ASTM D5334-08, ASTM E1225-09, and ASTM E1461-07, respectively. [6-8] Other related methods include IEEE Standard 442, ASTM D5930-09, and ASTM E2585-09. [9-11] These methods assume a static material, or in the case of a porous material, both a static solid matrix and static interstitial fluid. When a porous material has fluid flow, these methods fail. For the transient line source method, the failure causes overestimation of the thermal conductivity.

This paper develops a single probe transient moving line source method that calculates thermal and flow properties of a porous material from thermal data. The method extracts an effective thermal conductivity and thermal diffusivity for the combined solid and liquid thermal effects in the porous material, and the superficial flow velocity of the flowing fluid by manipulation of the transient analytical solution for a moving line source of heat from [12]. Three independent equations are derived by differentiation of the moving line source solution. The first equation is the transient analytical solution for a moving line source. The second equation is the

derivative of the first equation with respect to the natural logarithm of scaled time. The third equation is the derivative with respect to scaled time of the second equation. A characteristic scaled time is obtained from the root of the third equation and is inversely proportional to the Peclet number. The Peclet number is a scaled parameter relating the advection of heat through the material from fluid flow to the diffusion of heat through conduction. Using the Peclet number along with the second and third equations and characteristic scaled time, the thermal conductivity, thermal diffusivity, and superficial flow velocity are determined. This paper presents the proposed technique in detail, followed by a laboratory scale experiment used to validate the technique. The method is then applied to field data from a standing column well used in ground source heat pump systems showing the applicability of the method.

DERIVATION OF SINGLE PROBE METHOD

In this section, the traditional transient line source method is presented and its application to extracting properties is analyzed. The proposed single probe transient moving line source method is then derived from the transient analytical solution for a moving line source from Woods and Ortega [12].

Transient Line Source Method

The traditional transient line source method is derived using Green's functions for the linear operator given by

$$\frac{1}{\alpha} \frac{\partial T}{\partial t} = \nabla^2 T \quad (1)$$

Heat is generated by a line source in the spatial domain producing a temperature distribution. The temperature distribution is modeled by the exponential integral function from Carslaw and Jaeger [13] for a stationary continuous line source of heat given by

$$T_p - T_\infty = \frac{Q}{4\pi Hk} \left[\ln \left(\frac{4\tau}{e^\gamma} \right) - \sum_{n=1}^{\infty} \frac{1}{n \cdot n!} \left(\frac{1}{4\tau} \right)^n \right] \quad (2)$$

Here $\tau = \alpha t / R^2$ and is the scaled time, α is the thermal diffusivity of the material, R is the radius of the probe, H is the height of the probe, T_p is the temperature of the probe, and k is the thermal conductivity of the material. At large times ($\tau \gg 1$), the summation approaches zero and the solution simplifies to a function of the logarithm of scaled time. Taking the derivative of Eq. (2) with respect to the natural logarithm of time yields a constant at large times given by

$$\frac{dT}{d \ln(t)} = \frac{Q}{4\pi Hk} \quad (3)$$

Rearranging Eq. (3) gives an explicit expression for thermal conductivity

$$k = \frac{Q}{4\pi H \frac{dT}{d \ln(t)}} \quad (4)$$

Inserting the thermal conductivity into Eq. (2) and rearranging yields an explicit form for thermal diffusivity evaluated at large times given by

$$\alpha = \frac{e^\gamma R^2}{4t} \exp \left(\frac{4\pi Hk (T_p - T_\infty)}{Q} \right) \quad (5)$$

Transient Moving Line Source Method

When an advection term is present due to fluid flow through the porous material, the governing equation becomes

$$\frac{1}{\alpha} \frac{\partial T}{\partial t} + \frac{(\rho C_p)_f}{k} U \frac{\partial T}{\partial x} = \nabla^2 T \quad (6)$$

Here $(\rho C_p)_f$ is the volumetric heat capacity of the flowing fluid. The governing equation is solved by Woods and Ortega [12] as a moving line source problem with a solution for the probe temperature given by

$$T_p - T_\infty = \frac{Q}{4\pi Hk} I_0 \left(\frac{Pe}{2} \right) \int_0^\tau \frac{\exp \left(-\frac{Pe^2 s}{4} - \frac{1}{4s} \right)}{s} ds \quad (7)$$

Here Pe is the Peclet number given by

$$Pe = \frac{(\rho C_p)_f U R}{k} \quad (8)$$

The probe temperature in Eq. (7) can be scaled as

$$\theta = \frac{T - T_\infty}{\frac{Q}{2\pi Hk}} \quad (9)$$

and thus

$$\theta_p = \frac{1}{2} I_0 \left(\frac{Pe}{2} \right) \int_0^\tau \frac{\exp \left(-\frac{Pe^2 s}{4} - \frac{1}{4s} \right)}{s} ds \quad (10)$$

The solution is presented in Fig. 1 for Peclet numbers of 0, 0.25, 0.5, 0.75, and 1; the semi-logarithmic plot is shown in Fig. 2. The response for $Pe = 0$ asymptotes to a log-linear behavior as seen in Fig. 2. The slope of this log-linear behavior is 0.5, as expected from the non-dimensional form of Eq. (2). For $Pe > 0$, the response asymptotes to a constant value at large τ ; this large τ asymptote derived from Eq. (10) is given by

$$\theta_p = I_0 \left(\frac{Pe}{2} \right) K_0 \left(\frac{Pe}{2} \right) \quad (11)$$

Now, differentiating Eq. (10) with respect to the natural logarithm of τ gives

$$\frac{d\theta_p}{d\ln(\tau)} = \frac{1}{2} I_0\left(\frac{Pe}{2}\right) \exp\left(-\frac{Pe^2\tau}{4} - \frac{1}{4\tau}\right) \quad (12)$$

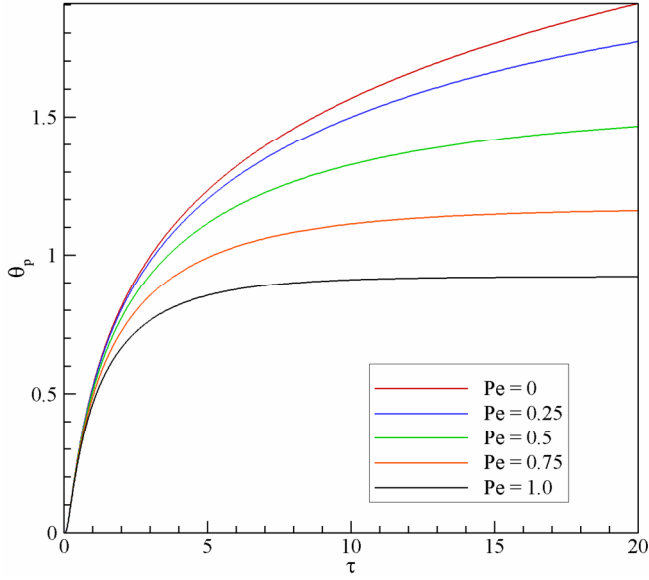


Fig. 1: Scaled temperature versus scaled time for varying Peclet numbers

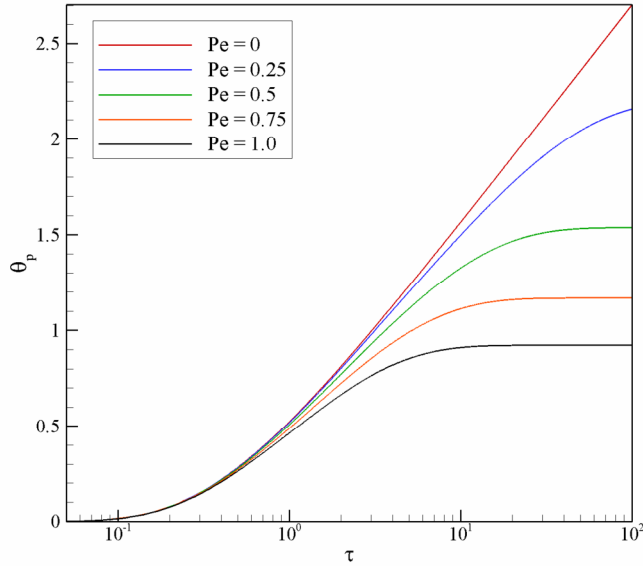


Fig. 2: Semi-logarithmic plot of scaled temperature versus scaled time.

And finally, differentiating Eq. (12) with respect to τ becomes

$$\frac{d}{d\tau} \left(\frac{d\theta_p}{d\ln(\tau)} \right) = \frac{1}{8} I_0\left(\frac{Pe}{2}\right) \exp\left(-\frac{Pe^2\tau}{4} - \frac{1}{4\tau}\right) \left(\frac{1}{\tau^2} - Pe^2 \right) \quad (13)$$

Taking the root of Eq. (13), a characteristic scaled time, τ_c , is obtained which is the point of inflection of the temperature

response curves in Fig. 2. Remarkably, the characteristic time is given by the simple expression

$$\tau_c = \frac{1}{Pe} \quad (14)$$

This point of inflection represents the transition from a diffusively dominated response to one in which both diffusion and advection play a role in heat removal from the probe. Figure 3 shows Eq. (12) plotted at various Pe . The characteristic time τ_c given by Eq. (14) is the time at which the maxima are achieved in Fig. 3.

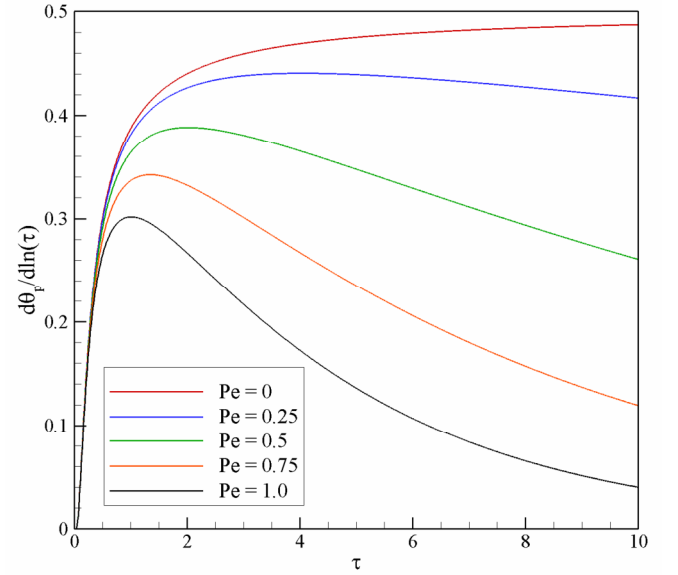


Fig. 3: The semi-logarithmic derivative versus scaled time.

Evaluating Eqs. (10) and (12) at τ_c and dividing Eq. (12) by Eq. (10) produces a function that is solely dependent on Peclet number

$$f(Pe) = \frac{\left. \frac{d\theta_p}{d\ln(\tau)} \right|_{\tau_c}}{\theta_p|_{\tau_c}} = \frac{\exp\left(-\frac{Pe}{2}\right)}{\int_0^{1/Pe} \frac{\exp\left(-\frac{Pe^2 s}{4} - \frac{1}{4s}\right)}{s} ds} \quad (15)$$

Since $d\ln(\tau) = d\ln(t)$, the function of Pe in Eq. (15) may be written as

$$f(Pe) = \frac{\left. \frac{dT_p}{d\ln(t)} \right|_{t_c}}{T_p|_{t_c} - T_\infty} \quad (16)$$

Here t_c is the characteristic time from Eq. (14) in dimensional form. The values of $dT_p/d\ln(t)$ and $T_p - T_\infty$ at t_c , can be extracted from empirical data by analyzing the semi-logarithmic plot of the empirical data at the point of inflection.

Therefore, $f(Pe)$ may be obtained from the empirical data, and hence, the Peclet number may be calculated using Eq. (15).

To simplify the analysis, a correlation for Eq. (15) was calculated for Peclet numbers of 0.01 to 5 and given by the polynomial

$$Pe = 3.4056(f(Pe))^2 - 0.6567f(Pe) - 0.0289 \quad (17)$$

Once the Peclet number is known, the thermal conductivity is calculated by rearranging Eq. (12) and evaluating at τ_c producing

$$k = \frac{Q}{4\pi H \frac{dT_p}{d \ln(t)} \bigg|_{t_c}} I_0 \left(\frac{Pe}{2} \right) \exp \left(-\frac{Pe}{2} \right) \quad (18)$$

The thermal diffusivity is calculated using the definition of τ_c and Eq. (14) given by

$$\alpha = \frac{1}{Pe} \left(\frac{R^2}{t_c} \right) \quad (19)$$

The superficial flow velocity is then obtained by rearranging Eq. (8) yielding

$$U = Pe \left(\frac{k}{(\rho C_p)_f R} \right) \quad (20)$$

Typically $(\rho C_p)_f$ can be accurately estimated from thermal property data. For example, in standing column wells for ground source heat pumps, $(\rho C_p)_f$ is estimated using the thermal properties of water.

EMPIRICAL DATA ACQUISITION

Validation Experiment

Empirical data was acquired from a laboratory scale test apparatus shown in Fig. 4. The laboratory test apparatus was a 0.2m x 0.36m x 0.1m (L x W x H) block of paver sand with a cartridge heater (0.003175m radius x 0.1m height) acting as the line source positioned in the geometric center of the sand. The heat source contained an embedded type K thermocouple which provided the probe temperature. The temperature was referenced to an ice point for accuracy. Water was pumped from one reservoir through the length of the sand to another reservoir. The flow was controlled with a ball valve and flow rate was calculated with a graduated cylinder and stopwatch at the start and end of each test.

Three tests were performed on the laboratory scale apparatus. The parameters of each test are shown in Table 1.

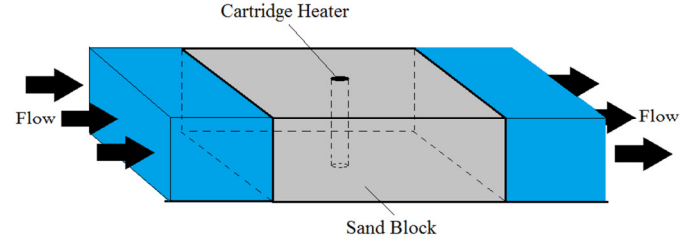


Fig. 4: Laboratory Scale Apparatus Diagram.

Table 1: Operational Parameters of Laboratory Scale Tests

	Test #1	Test #2	Test #3
Flow (ml/s)	0	3.03	2.38
Heat Load (W)	5.025	2.23	3.24

Field Experiment

Empirical data from a field test was obtained for a standing column well used in a geothermal heat pump system [14]. A diagram of the field test apparatus is shown in Fig. 5. Water was pumped down the annulus of a standing column well where it thermally interacted with the ground formation around it. It then returned through a pipe located in the center of the well to the top where it was heated and reinjected. A full analysis of standing column well physics is presented by Woods and Ortega [12]. The field test specifications were: $Q = 58184.2W$, $H = 153m$, $m_w = 2.62kg/s$, and $R = 0.0889m$. The superficial flow velocity and thermal properties were unknown.

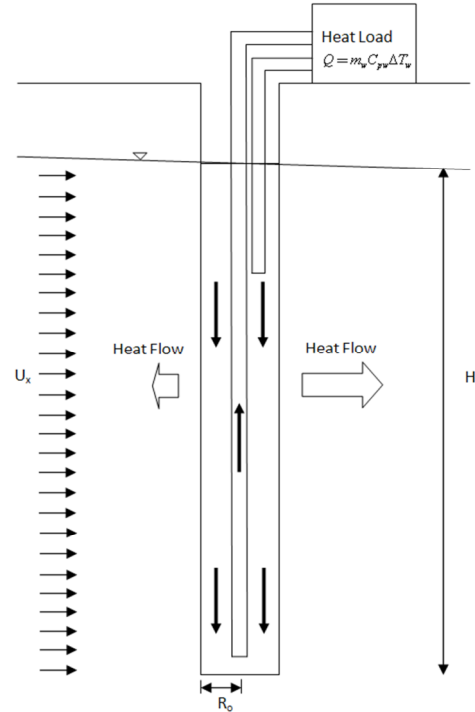


Fig. 5: Diagram of the geothermal well in the field.

RESULTS AND DISCUSSION

In this section, the empirical data are presented along with the results from application of the transient moving line source method on the empirical data for determining the thermal conductivity, thermal diffusivity, and superficial flow velocity. The empirical data from Test #1 are evaluated using the transient line source method for no flow conditions. This provides the baseline properties to compare with Test #2 and Test #3 which are evaluated with the transient moving line source method. The field test data results are presented showing the comparison of the calculated values from the transient line source method to the transient moving line source method.

Empirical Data Acquisition Results

The results for the laboratory scale tests are presented in Figs. 6 and 7 for normal and semi-logarithmic time scales. The results from the field test are presented in Fig. 8. All results are in dimensional units because the scaling parameters require known thermal properties which are unknown for the data. The field data is manipulated by the method derived by Woods and Ortega [12] for analyzing standing column well data. As expected, Test #1 in Fig. 7 follows the profile of $Pe = 0$ from Fig. 2 with a log-linear profile at large times. The transient line source method is applied to Test #1 to extract the values of thermal conductivity and thermal diffusivity for the laboratory apparatus. These thermal property values are the baseline for comparison with all Tests #2 and #3 data. The failure of the transient line source method, when flow conditions exist, is shown by applying it to Test #2, Test #3, and the field test data. The transient moving line source method is applied to Test #2, Test #3, and the field test data to estimate the thermal conductivity, thermal diffusivity, and superficial flow velocity for each set of data. All of the results are shown in Table 2.

The traditional transient line source method showed 133% and 244% error in the thermal conductivity calculation for Test #2 and Test #3, respectively compared to the baseline Test #1 results. The thermal diffusivity comparison showed 54% and 69% error, respectively.

The proposed transient moving line source method showed 1% and 4% error in the thermal conductivity calculation for Test #2 and Test #3, respectively, and 8% and 1% error for thermal diffusivity, respectively, as compared to the baseline Test #1 results. The superficial flow velocity error is 34% and 43% for Tests #2 and #3, respectively. These errors arise from fluctuations in the data due to environmental and instrumentation errors, which are amplified upon differentiation. As an example, Figs. 9 and 10 show the equivalent plot from Fig. 3 for the empirical data from Test #2

and the field test. Current work is ongoing to process the data in such a way as to reduce the propagation of errors.

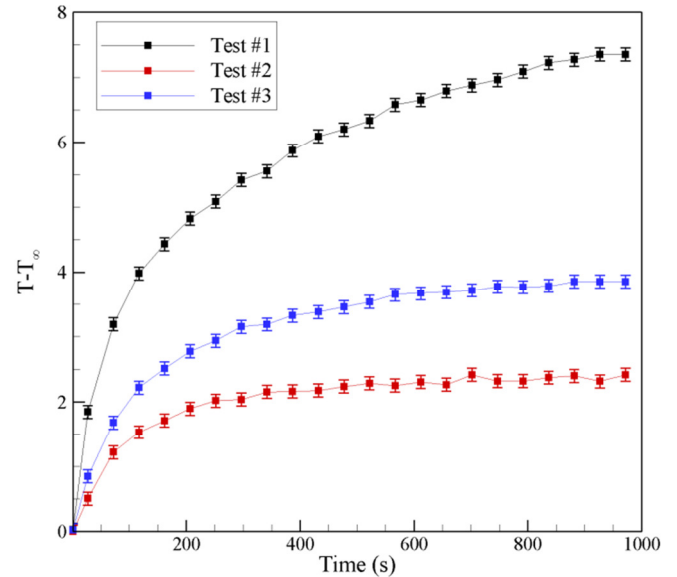


Fig. 6: Temperature response of the laboratory tests.

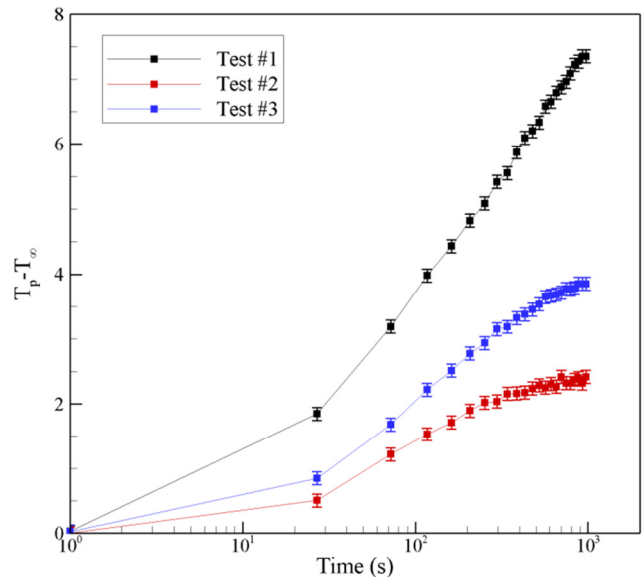


Fig. 7: Temperature response of laboratory tests in the semi-logarithmic time domain.

When calculating the property values in Table 2, a point was taken from each set of data estimated to be the point of inflection of the semi-logarithmic plot. The point of inflection was estimated using Fig. 7 to be around 100 seconds for both tests. The actual times used were 94.5s and 142.5s for Tests #2, and #3, respectively.

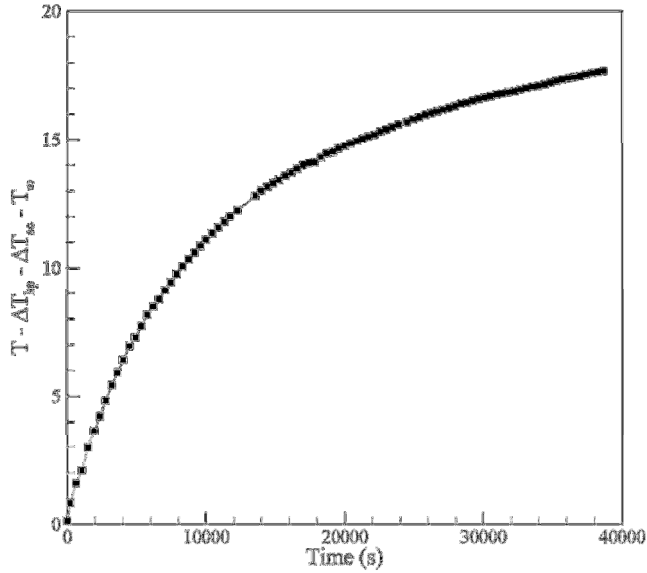


Fig. 8: Field test data analyzed using the method from Woods and Ortega [12].

Table 2: Results from property extraction methods (*denotes that the method does not calculate the property, **denotes that the data was not obtained during the test)

	Thermal Conductivity (W/mK)	Thermal Diffusivity (m ² /s)	Superficial Flow Velocity (m ³ /m ² s)	Measured Superficial Flow Velocity
Measured Using the Transient Line Source Method				
Test #1	2.36	3.76E-07	*	0
Test #2	5.51	1.73E-07	*	8.10E-05
Test #3	8.11	1.15E-07	*	6.36E-05
Field Data	6.81	4.68E-06	*	**
Measured Using the Transient Moving Line Source Method				
Test #2	2.39	3.60E-07	5.35E-05	8.10E-05
Test #3	2.55	3.74E-07	3.63E-05	6.36E-05
Field Data	4.32	1.70E-06	6.01E-06	**

Although the property values are not exact, the values provide an estimate of the actual values that may be obtained by plotting Eq. (7) with respect to the actual data and varying the properties around the estimated values until a reasonable fit is achieved. Figure 11 shows the analytical solution from Eq. (7) superimposed over the data for the measured values of the properties in the laboratory tests. Figure 12 shows the analytical solution from Eq. (7) using the calculated values of the properties from the field test, without modification, superimposed over the field data.

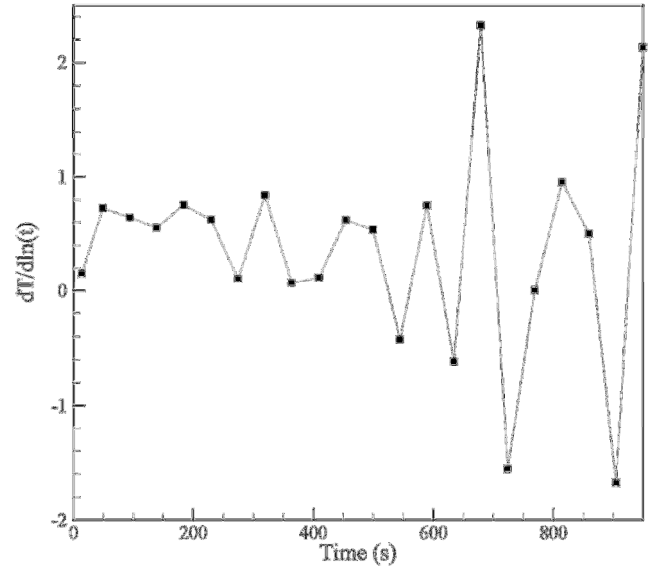


Fig. 9: Derivative of temperature with respect to the natural logarithm of time for data from Test #2.

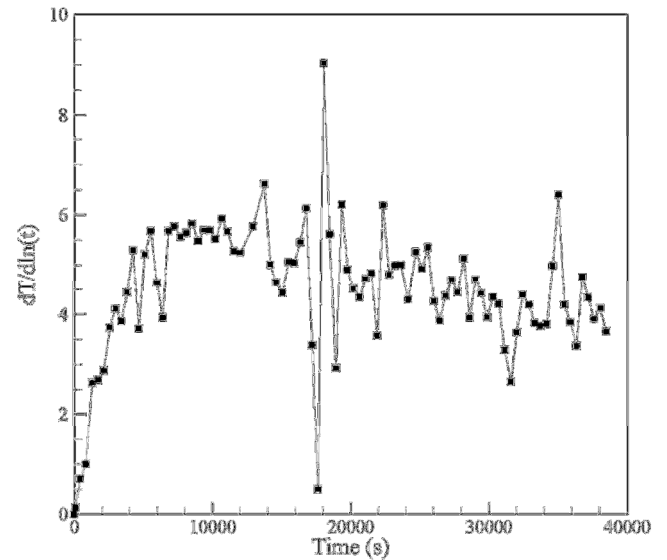


Fig. 10: Derivative of temperature with respect to the natural logarithm of time for the field data.

CONCLUSIONS

A single probe transient moving line source method was developed to obtain the thermal conductivity, thermal diffusivity, and superficial flow velocity in a porous material from empirical data. The method was derived analytically from the transient analytical solution for a moving line source of heat and compared to the analytics of the traditional transient line source method.

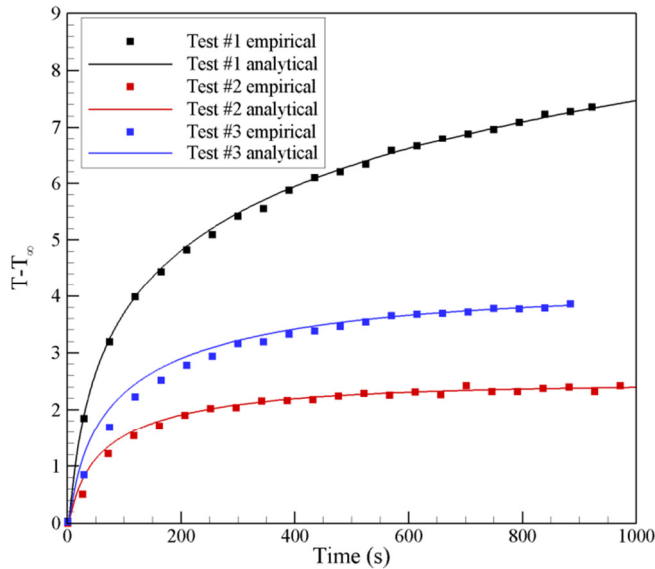


Fig. 11: Comparison of empirical data to the analytical function using measured property values.

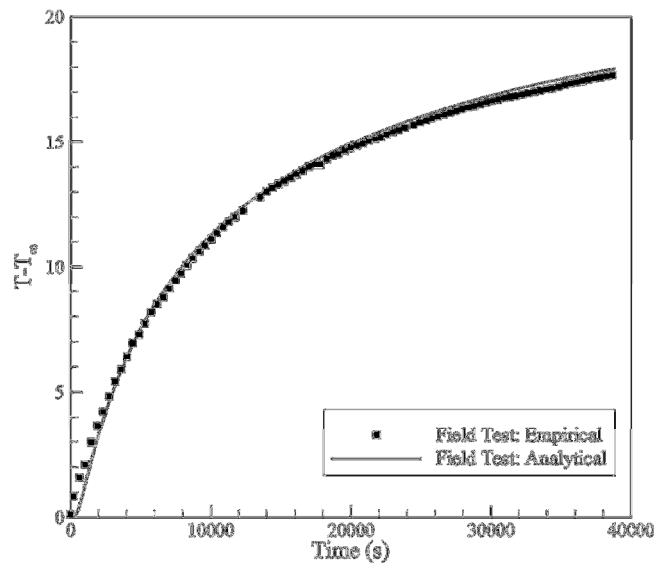


Fig. 12: Comparison of empirical data to the analytical function using the calculated properties in Table 2.

Empirical data were obtained from a laboratory scale apparatus. An initial test contained no flow and was used to extract the actual property values of the material using the traditional line source method. The second and third tests operating with flow were analyzed with the proposed transient moving line source method and the extracted property values were compared to the values from the first test for validation of the method. The error was less than 8% for the thermal properties and ~40% for the superficial flow velocity. The traditional transient line source method was applied to the second and third tests to show the failure of the method. Errors

of 54-244% resulted for the thermal properties, and no superficial flow velocity could be calculated.

Specific conclusions from the work presented here are summarized as follows:

1. The transient moving line source method is a valid method for estimating thermal conductivity, thermal diffusivity, and superficial flow velocity from a porous material with flow conditions.
2. The traditional transient line source method fails when flow is present in the porous material.
3. The accuracy of the method is dependent on the fluctuation of the data, caused by instrumentation and environmental errors, which are amplified by differentiation in the method. Data reduction procedures are being explored to reduce the amplification of errors.
4. The implementation of the method does not require changing the traditional method of data acquisition but rather the traditional method of data reduction and analysis.

NOMENCLATURE

k	Thermal conductivity of material (Combined solid and fluid)
C_p	Heat capacity
H	Height of probe
Q	Heat load applied to probe
R	Radius of Probe
T	Temperature
U	Superficial flow velocity

Greek symbols

α	Thermal diffusivity
ΔT_{hp}	Temperature associated with external heat load for a geothermal well
ΔT_{sc}	Temperature associated with short circuiting of heat in a geothermal well
γ	Euler's constant
ρ	Density
θ	Scaled Temperature

Subscripts and Superscripts

c	Associated with the characteristic time for the point of inflection
f	Flowing fluid
p	Thermal probe
∞	Infinite domain

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